Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart is at **3**, **8**, **13 m** at **0**, **1**, **2 s**.
1. **Predict its position** at **7 s**.
2. **State something you** are treating as true and a measurement that could challenge it.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths.
Choose an origin and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at **2 m**, reaches **6 m**, then ends at **4 m**. Distance=**4**+**2**=**6 m**. Displacement=**4**-**2**=**2 m** right.

Your turn: The walker goes from the **2 m** floor mark to the **6 m** mark, then back to **4 m**. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. **Label each length**.

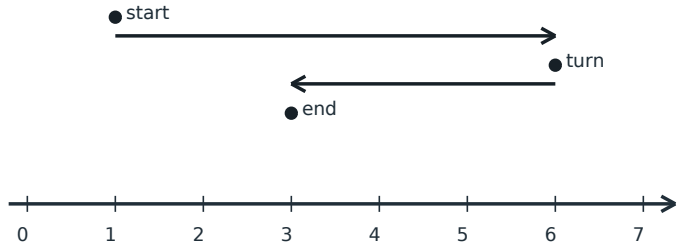


Position (m); right is positive

Marzano check - circle: ① ② ③

03 A walk along the hallway

You walk along a hallway marked in meters. Start at **1 m**, turn around at **6 m**, and stop at **3 m**. **Label the start**, turn and finish on the route.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Count the whole path

For the route in **3**, find each leg and the total distance.
First leg: ____ m. Return leg: ____ m. Total: ____ m.

Marzano check - circle: ① ② ③

05 Connect start to finish

For the same route, draw one arrow from start to end.
1. **Give the displacement** and final position.
2. Are they the same number?

Support: I end ____ m to the ____ of where I started.

Marzano check - circle: ① ② ③

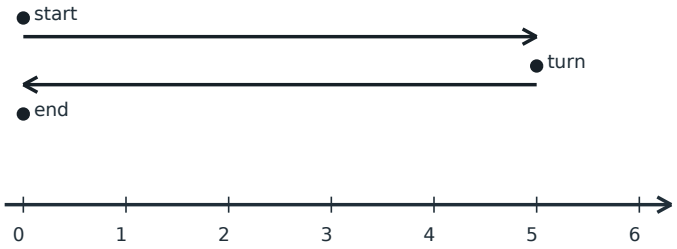
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Continue down each column.

06 A complete trip back

You leave a doorway, walk **5 m** *east*, then return along the same path.
1. **Find distance** and displacement.
2. **Use the route** picture to **explain why only one** is zero.



Position (m); right is positive

Marzano check - circle: ① ② ③

07 Fix the wording

A student says, “My displacement was **8 m** because I walked **8 m** altogether.” Must that be true? Use a short route or sketch to explain.

Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

A cart is observed at **3**, **8**, **13 m**, one second apart. **Record two patterns** and point to values that support each.

Time (s)	Position (m)
0	3
1	8
2	13

Marzano check - circle: ① ② ③

09 Exit check

A robot starts at the **2 m** mark, travels to **7 m**, then returns to **5 m**.
1. **Sketch the route**.
2. **Find total distance**, displacement with direction, and final position.



Position (m); right is positive

Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

1. Design two routes that start at **1 m** and end at **4 m**: one of distance **3 m** and one of distance **7 m**.
2. Draw and **check both**.



Position (m); right is positive

Marzano check - circle: ① ② ③